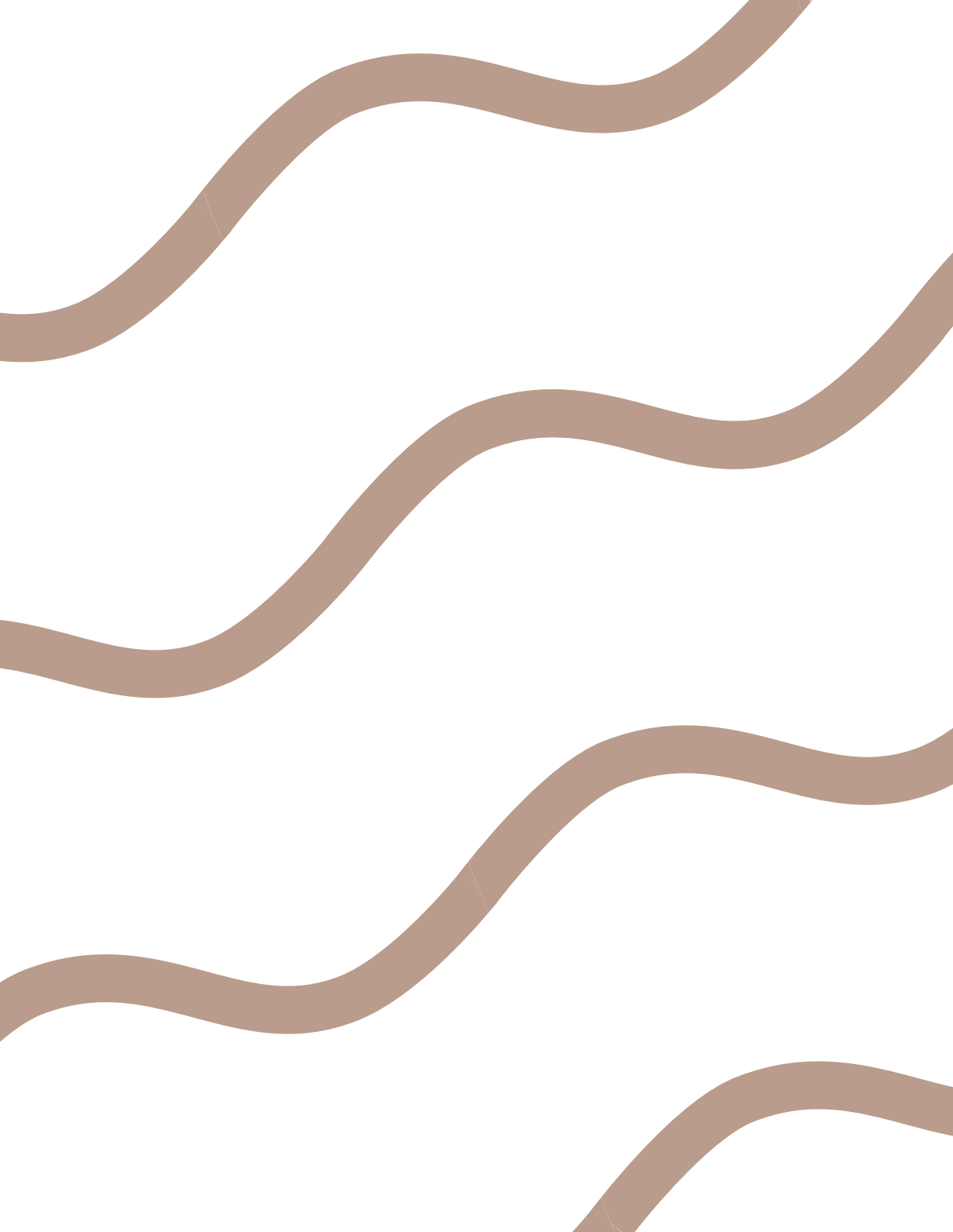




19. (本题满分 12 分)

已知函数 $y = y(x)$ 可导, 将区间 $[0, x]$ 上以 $y = y(x)$ 为曲边的曲边梯形面积记为 S , 在 $[0, x]$ 上的弧长为 aS ($a > 0$), 且 $y(0) = \frac{1}{a}$, 求曲线 $y = y(x)$ 的表达式.

(18) 设上凸曲线 $y = y(x)$ ($y > 0$) 上任一点 $M(x, y)$ 处的切线与 x 轴交于点 N , 且满足 $|OM| = |ON|$, $y(0) = 1, y'(x) > 0$, 求 $y = y(x)$.

$$a \int_0^x |y(t)| dt = \int_0^x \sqrt{1+[y'(t)]^2} dt,$$

$$y(0) = 1/a.$$

等式两边求导,得

$$a |y(x)| = \sqrt{1+[y'(x)]^2},$$

于是,有

$$y' = \pm \sqrt{a^2 y^2 - 1}.$$

对于 $y' = \sqrt{a^2 y^2 - 1}$, 有

$$\begin{aligned} \frac{dy}{\sqrt{a^2 y^2 - 1}} &= dx, \\ \int_{1/a}^y \frac{dy}{\sqrt{a^2 y^2 - 1}} &= \int_0^x dx, \\ \frac{1}{a} \ln(ay + \sqrt{a^2 y^2 - 1}) &= x. \end{aligned}$$

故

$$-\ln(ay + \sqrt{a^2 y^2 - 1}) = \ln(ay - \sqrt{a^2 y^2 - 1}) = -ax,$$

从而

$$\begin{cases} ay + \sqrt{a^2 y^2 - 1} = e^{ax}, \\ ay - \sqrt{a^2 y^2 - 1} = e^{-ax}. \end{cases}$$

两式相加得

$$y = \frac{e^{ax} + e^{-ax}}{2a}.$$

(18) 解 依题设, 凸曲线 $y = y(x)$, $y'(x) > 0$, 过 M 作 x 轴的垂线 MP , 如图 6-2 所示, 则

$$\frac{dy}{dx} = \frac{|MP|}{|NP|}, \quad |MP| = y, \quad |NP| = \sqrt{x^2 + y^2} + x$$

故 $y(x)$ 满足的微分方程为

$$\begin{cases} \frac{dy}{dx} = \frac{y}{x + \sqrt{x^2 + y^2}}, \\ y(0) = 1, \end{cases}$$

变形为

$$\frac{dx}{dy} = \frac{x + \sqrt{x^2 + y^2}}{y} = \frac{x}{y} + \sqrt{1 + \left(\frac{x}{y}\right)^2}.$$

令 $\frac{x}{y} = u$, 代入上式, 得 $u + y \frac{du}{dy} = u + \sqrt{1+u^2}$, 即 $\frac{du}{\sqrt{1+u^2}} = \frac{dy}{y}$, 两边积分, 得

$$\ln(u + \sqrt{1+u^2}) = \ln y + C_1,$$

故 $u + \sqrt{1+u^2} = C_2 y$ ($C_2 = e^{C_1} > 0$), 将 $\frac{x}{y} = u$ 代入并化简, 得 $1 = C_2^2 y^2 - 2C_2 x$, 由 $y(0) = 1$, 得 $C_2 = 1$.

所求 $y(x) = \sqrt{1+2x}$ ($x > -\frac{1}{2}$).

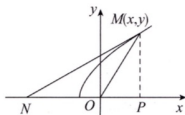


图 6-2